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1. Introduction

Network traffic analysis is a fundamental practice in cybersecurity used to monitor, inspect, and understand data communication between systems over a network. Every action performed on the internet, such as visiting websites, sending requests, or resolving domain names, generates network packets that can be captured and analyzed.

In this project, network traffic was captured and analyzed using Wireshark to understand how different protocols such as DNS, TCP, HTTP, and TLS operate in real-time communication. The main objective of this analysis is to identify how data is transmitted and to evaluate potential security risks such as unencrypted communication and information exposure.

This project demonstrates practical knowledge of network monitoring and provides insights into how cybersecurity professionals analyze traffic to detect vulnerabilities and suspicious behavior.

2. Tools Used

The following tools and technologies were used in this project:

- Wireshark for packet capture and analysis
- Web browser (Chrome/Firefox) to generate real network traffic
- Command Prompt / Terminal for network commands such as ping and nslookup
- Internet connection for real-time traffic generation

3. Methodology

The project was conducted using a structured approach to capture and analyze network traffic:

First, Wireshark was installed and executed with administrative privileges to allow access to network interfaces. The active network interface (Wi-Fi) was selected to begin capturing live traffic.

After starting the capture, real network activity was generated by:

- Visiting websites through a web browser
- Performing DNS lookups using the nslookup command
- Sending ICMP requests using the ping command

The captured traffic was then filtered using Wireshark display filters such as:

- DNS filter for domain resolution analysis
- TCP filter for connection behavior analysis
- TLS filter for encrypted communication analysis

This approach allowed a detailed inspection of different network layers and protocols.

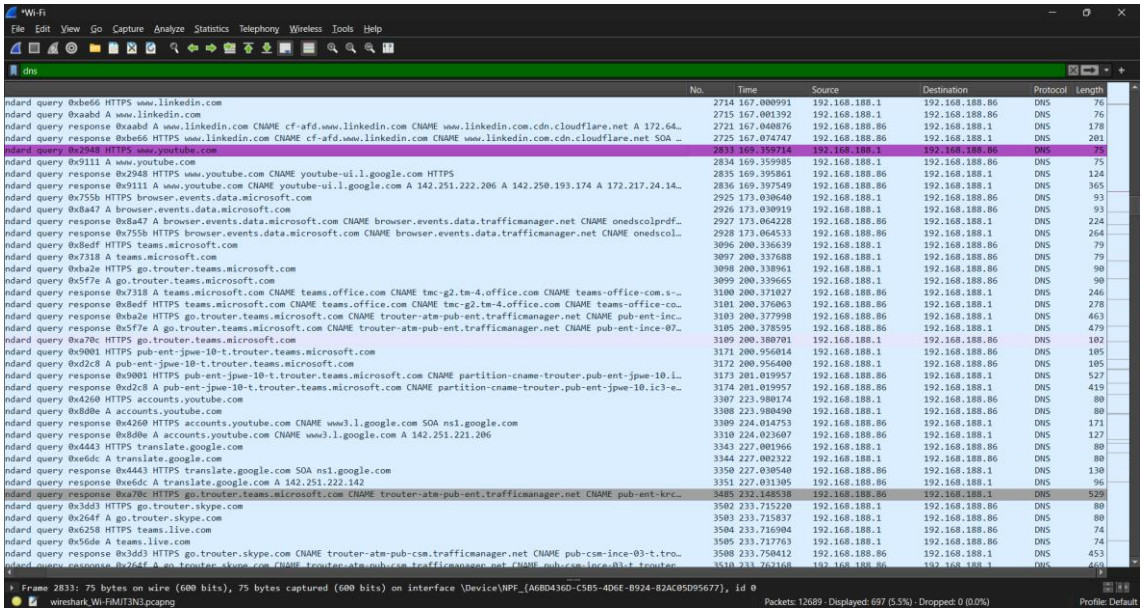
4. Analysis and Findings

4.1 DNS Analysis

DNS packets were observed during domain name resolution processes. The captured data showed queries and responses where domain names were translated into IP addresses.

This analysis revealed how user requests are translated into network-level communication and demonstrated that DNS queries can expose browsing activity.

Figure 1.Dns Queriers

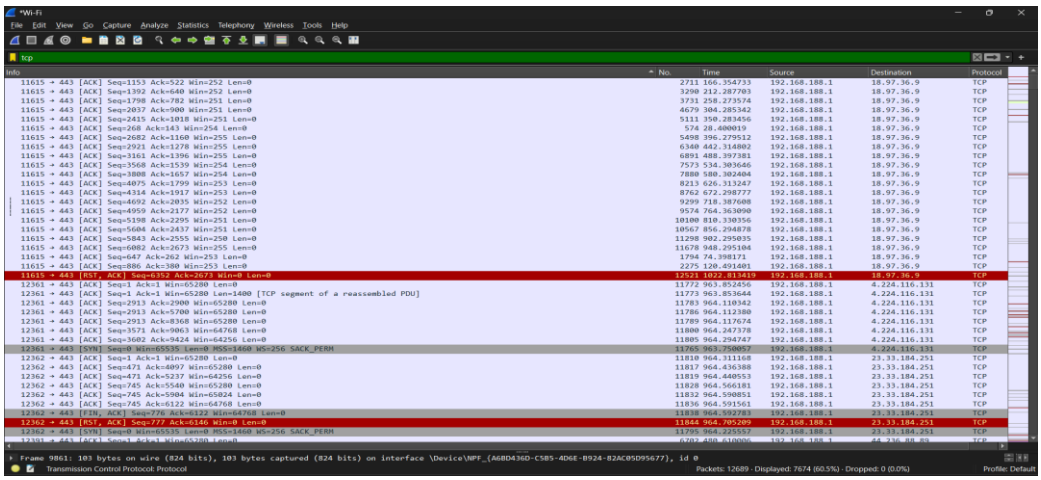


4.2 TCP Analysis

TCP traffic analysis showed the establishment of connections using the three-way handshake process: SYN, SYN-ACK, and ACK.

This confirms how reliable communication is established between client and server before data transfer begins.

Figure 2.TCP Analysis

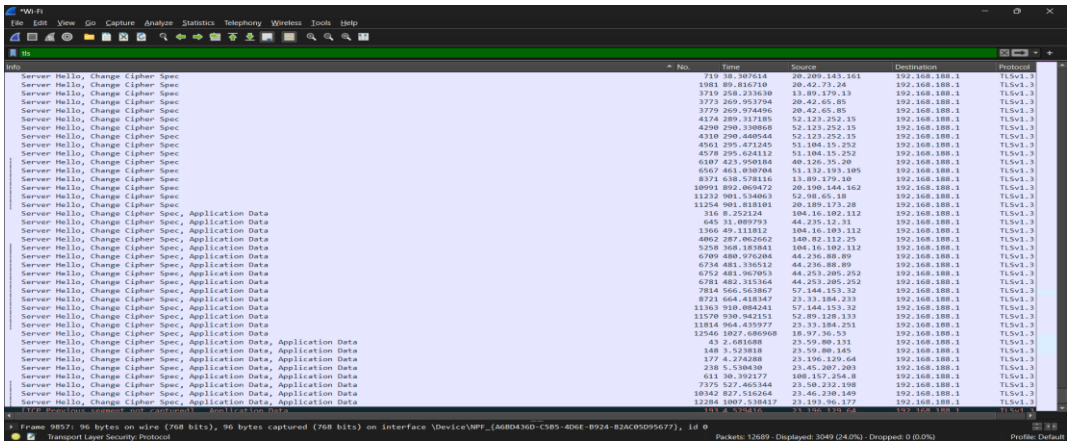


4.3 TLS Analysis

TLS packets were observed during secure communication with websites. The handshake process included Client Hello and Server Hello messages.

This indicates that most modern web traffic is encrypted, ensuring confidentiality and protecting data from interception.

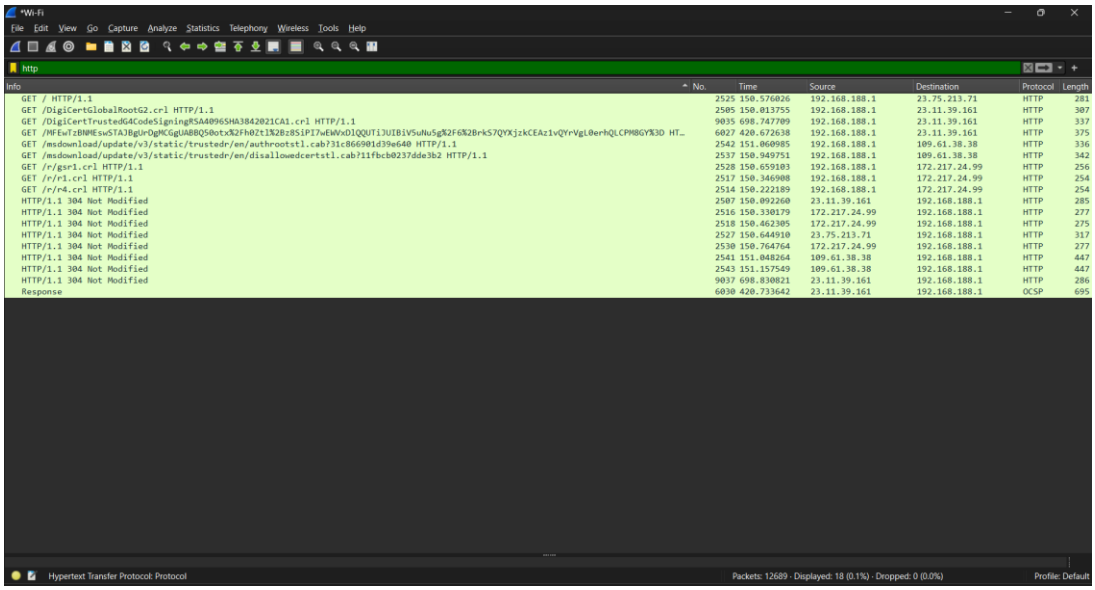
Figure 3.TSL Analysis



4.4 HTTP Analysis (Limited)

Minimal or no HTTP traffic was observed, as most websites now use HTTPS instead of unencrypted HTTP. This reflects modern security practices where encryption is enforced by default.

Figure 4.HTTP Analysis



5. Security Risks Identified

The following security considerations were identified during the analysis:

- DNS traffic can expose user browsing activity and visited domains
- Unencrypted HTTP traffic (if present) can expose sensitive information
- Network traffic patterns can be analyzed to understand user behavior
- Attackers may exploit unencrypted or misconfigured communication channels

6. Recommendations

Based on the findings, the following security improvements are recommended:

- Enforce HTTPS across all web communications to ensure encryption
- Implement DNS over HTTPS (DoH) to protect DNS queries
- Monitor network traffic using intrusion detection or SIEM tools
- Use secure network configurations to minimize exposure of sensitive data
- Educate users about secure browsing practices

7. Conclusion

This project successfully demonstrated the use of Wireshark for network traffic analysis. By examining DNS, TCP, and TLS traffic, it was possible to understand how data flows across networks and how modern encryption protects sensitive information.

The analysis highlights the importance of network security practices and demonstrates how cybersecurity professionals investigate and monitor network behavior to identify risks and vulnerabilities.